

Rates of Change & Concavity



Average rate of change

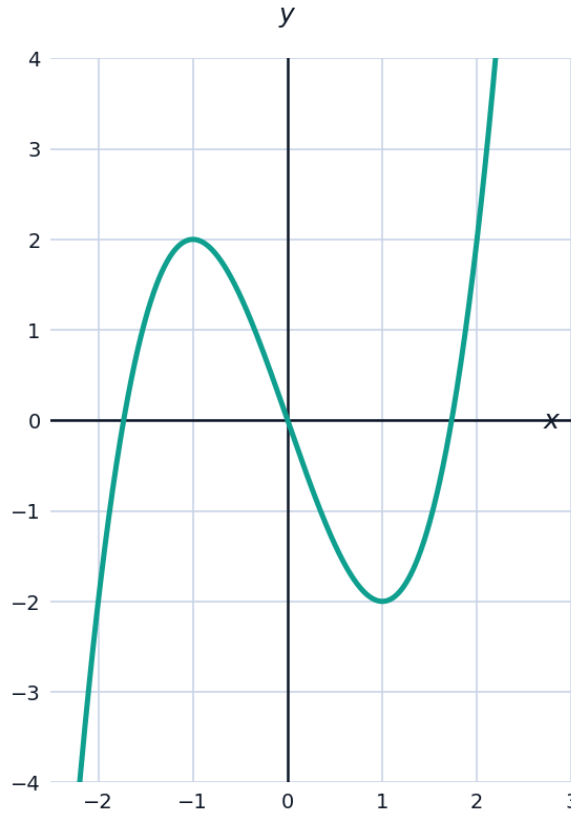
- 1 Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.
 - 2 A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.
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Increasing/decreasing behavior from rate of change

- 3 A function f has average rates of change of 2 on $[0, 1]$, 5 on $[1, 2]$, and 9 on $[2, 3]$. Determine whether f appears to be concave up or concave down on $[0, 3]$, and explain.
 - 4 A different function g has average rates of change of 8 on $[0, 1]$, 5 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of g on $[0, 3]$.
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Visualizing concavity and inflection

- 5 The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.



- a Describe the concavity of the graph on $(-2, 0)$.
- b Describe the concavity of the graph on $(0, 2)$.
- c State what kind of point $x = 0$ represents.

Rate of change of the rate of change

- 6 Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.

Applying rate-of-change reasoning to models

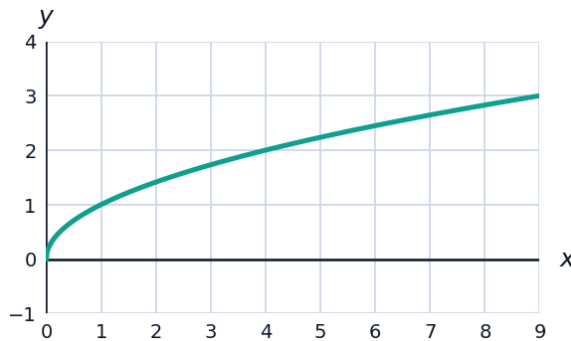
- 7 A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \leq t \leq 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.
- 8 For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as t increases past 4.

Estimating rate of change from a table

- 9 A car's odometer reading (miles) is recorded every hour: $t = 0$: 120; $t = 1$: 175; $t = 2$: 238; $t = 3$: 310. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.
- 10 Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.

Visualizing rate of change on a graph

- 11 The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as x increases.



- a Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$.
- b Explain how the shrinking steepness in the graph relates to your two AROC values.
- c State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern.

Concavity and real-world interpretation

- 12 A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.